

**SIMATS ENGINEERING
ASSESSMENT TOOL 3**

COURSE CODE: CSA5117 COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO4: *Implement best security practices for IP, email, and web service using PGP, S/MIME for enhancing email Security.CO (BL3, BL6)*

Assessment Tool Weightage: 10%

QUESTIONS: 5

TOTAL MARKS: 50

**Annexure A
Comparative Analysis**

Code of Conduct:

I CH.Bhavya Sri/192411249____(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

CH.Bhavya Sri
Signature of the student:

Annexure A
Comparative Analysis

1. Compare Authentication Header (AH) and Encapsulating Security Payload (ESP) in IPSec with respect to security services, performance overhead, and suitability for different network scenarios. Analyze which is more appropriate for a VPN deployment.
2. Compare transport mode and tunnel mode in IPSec architecture. Evaluate their effectiveness in securing host-to-host versus network-to-network communication.
3. Perform a comparative analysis of Pretty Good Privacy (PGP) and S/MIME in terms of key management, trust models, scalability, and enterprise usability. Recommend the most suitable approach for a corporate email system.
4. Compare rule-based and machine learning-based email spam detection techniques. Analyze their effectiveness in handling evolving spam patterns and minimizing false positives.
5. Compare manual key management and automated key management (e.g., using IKE) in IPSec. Evaluate their impact on security, scalability, and operational complexity in large organizations.

1. Compare AH and ESP in IPSec

Authentication Header (AH) and Encapsulating Security Payload (ESP) are two protocols used by IPSec to secure communication over an IP network. AH mainly provides **authentication and integrity**, whereas ESP can provide **confidentiality, integrity, and authentication**.

Comparison of AH and ESP

Feature	AH	ESP
Authentication	Yes	Yes
Data integrity	Yes	Yes
Confidentiality	No	Yes
Encryption	No	Yes
Overhead	Lower	Higher
Main purpose	Integrity and authentication	Encryption, integrity and authentication
Suitable for	Non-confidential traffic	Confidential traffic and VPNs

Explanation

Authentication Header (AH):

AH ensures that the packet has not been modified during transmission and helps authenticate the source of the packet. However, AH **does not encrypt the data**, so the contents can still be read if intercepted.

Encapsulating Security Payload (ESP):

ESP can encrypt the packet payload, making the information unreadable to unauthorized users. It can also provide integrity and authentication. Because encryption requires additional processing, ESP generally has more overhead than AH.

Which is better for VPN?

For a VPN, **ESP is generally more appropriate**. VPNs are commonly used to carry confidential organizational information over public networks. ESP provides encryption in addition to integrity and authentication, making it suitable for protecting sensitive traffic.

Conclusion

AH is useful when authentication and integrity are required without encryption. **ESP is preferred for most VPN deployments because it provides confidentiality along with integrity and authentication.**

2. Compare Transport Mode and Tunnel Mode in IPSec

IPSec provides two modes of operation: **Transport Mode** and **Tunnel Mode**. The main difference is the amount of the IP packet that is protected.

Comparison

Feature	Transport Mode	Tunnel Mode
Protection	Protects the IP payload	Protects the complete original IP packet
Original IP header	Remains visible	Encapsulated inside new packet
Overhead	Lower	Higher
Typical use	Host-to-host	Network-to-network
VPN use	Less common	Very common
Security	Good	Stronger protection for complete packet

Explanation

Transport Mode:

In transport mode, the original IP header remains in the packet, while the payload is protected by IPSec.

Structure:

IP Header → IPSec Header → Protected Data

It is mainly suitable when **one host communicates directly with another host**. Because only the payload is protected, it creates less overhead.

Tunnel Mode:

In tunnel mode, the entire original IP packet is protected and placed inside a new IP packet.

Structure:

New IP Header → IPSec Header → Original IP Header + Data

Tunnel mode is especially useful when **two networks communicate through VPN gateways**. The original packet is hidden inside the protected tunnel.

Host-to-host vs Network-to-network

For **host-to-host communication**, transport mode is generally suitable because it has lower overhead.

For **network-to-network communication**, tunnel mode is more suitable because it protects the complete original packet and works well with VPN gateways.

Conclusion

Transport Mode → Host-to-Host

Tunnel Mode → Network-to-Network / VPN

Therefore, tunnel mode is generally preferred for connecting branch offices through a VPN.

3. Compare PGP and S/MIME

PGP and S/MIME are technologies used to provide secure email communication through encryption and digital signatures. The assessment specifically requires comparison based on **key management, trust models, scalability, and enterprise usability**.

Comparison

Feature	PGP	S/MIME
Key management	User-controlled	Certificate-based
Trust model	Web of Trust	Certificate Authority
Scalability	Moderate	High
Centralized management	Limited	Strong
Encryption	Yes	Yes
Digital signatures	Yes	Yes
Enterprise usability	Moderate	High
Administration	More user involvement	Easier centralized administration

Explanation

PGP (Pretty Good Privacy):

PGP uses public-key cryptography for encryption and digital signatures. Its trust model is called the **Web of Trust**, where users can verify and trust other users' public keys.

This gives users more control, but managing keys can become difficult when an organization has a large number of employees.

S/MIME:

S/MIME also provides encryption and digital signatures, but it uses **digital certificates** issued through trusted Certificate Authorities (CAs). Because certificates can be centrally issued, renewed, and revoked, S/MIME is easier to manage in a corporate environment.

Scalability

- PGP can work well for smaller or decentralized groups, but key management becomes more complicated as the number of users increases.
- S/MIME is more suitable for large organizations because certificate management can be centrally controlled.

Recommendation

For a **corporate email system**, **S/MIME is the more suitable choice** because it provides centralized certificate management, better enterprise administration, and good scalability.

Conclusion

PGP is useful when decentralized key management is preferred. **S/MIME is generally better for corporate environments where centralized management and scalability are important.**

4. Compare Rule-Based and Machine Learning-Based Spam Detection

Email spam detection is important because organizations face increasing spam and phishing attacks. The assessment asks for comparison of **rule-based and machine-learning-based detection** and their ability to handle changing spam patterns and false positives.

Comparison

Feature	Rule-Based Detection	Machine Learning Detection
Method	Predefined rules	Learns patterns from data
Adaptability	Low	High
New spam patterns	Difficult to detect	Better at detecting new patterns
Maintenance	Manual updates	Requires training and monitoring
Complexity	Low	Higher
False positives	Can be high	Can be reduced with good training
Processing	Generally simple	More computationally demanding
Suitable for	Known spam patterns	Large and changing spam patterns

Explanation

Rule-Based Detection:

A rule-based system uses predefined conditions to identify spam. For example, an email can be marked as suspicious if it contains certain words, suspicious links, or comes from a known spam source. The main advantage is that it is **simple and easy to understand**.

However, attackers continuously change their methods. If a new spam pattern does not match an existing rule, the system may fail to detect it.

Machine Learning-Based Detection:

Machine learning systems learn patterns from previously classified emails. They can analyze features from the email header and content and classify messages as legitimate, spam, or phishing.

Basic process:

Email → Feature Extraction → ML Model → Classification → Spam / Legitimate

Evolving spam patterns

Machine learning is generally more adaptable because it can be retrained using new examples. This makes it useful when attackers frequently change their techniques.

False positives

A **false positive** occurs when a legitimate email is incorrectly classified as spam.

False positives can be reduced by:

Using good-quality training data

Regularly updating the model

Monitoring classification results

Using a quarantine system

Allowing users to report incorrect classifications

Conclusion

Rule-based systems are simple and effective for **known threats**, while machine learning is more suitable for **changing and complex spam patterns**. A combination of both approaches can provide stronger email protection.

5. Compare Manual and Automated Key Management

IPSec requires proper management of cryptographic keys and Security Associations (SAs). The assessment asks for comparison of manual and automated key management, particularly using **IKE**, in terms of security, scalability, and operational complexity.

Comparison

Feature	Manual Key Management	Automated Key Management
Key configuration	Manually performed	Automatically negotiated
Human effort	High	Low
Scalability	Low	High
Key renewal	Manual	Automatic
Human errors	More likely	Reduced
Security consistency	Depends on administrator	More consistent
Large organizations	Difficult	Suitable
Operational complexity	High	Lower
Example	Manually configured keys	IKE

Explanation

Manual Key Management:

In manual key management, administrators configure encryption keys and security parameters on network devices themselves.

For a small network, this may be manageable. However, when an organization has many branch offices and devices, manually configuring every key becomes time-consuming.

It can also result in:

Expired keys

Mismatched keys

Configuration errors

Difficulty in key replacement

Automated Key Management:

Automated systems such as **IKE (Internet Key Exchange)** allow devices to negotiate security parameters and cryptographic keys automatically.

Basic IKE process

Device A

↓

IKE Negotiation

↓

Authentication

↓

Key Exchange

↓

Security Association Creation



Secure IPSec Communication

This reduces the amount of manual configuration required.

Security and scalability

Automated key management is more suitable for large organizations because keys can be renewed according to predefined policies. It also reduces human errors and makes it easier to manage many IPSec connections.

Operational complexity

Although automated key management requires proper initial configuration, it significantly reduces the ongoing administrative workload.

Conclusion

Manual key management is suitable mainly for small and simple environments, while automated key management using IKE is better for large organizations. Automation improves scalability, consistency, key renewal, and operational efficiency.